

Project Initialization and Planning Phase

Date	18 June 2025
Team ID	SWTXT_20251747173488
Project Title	Productivity Prediction Web App
Maximum Marks	3 Marks

Project Proposal: Productivity Prediction Web App

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Introduction

This project proposal outlines a machine learning-based solution to predict garment worker productivity, addressing inefficiencies in the textile industry. Using the "Productivity Prediction of Garment Employees" dataset, the proposed web app employs Linear Regression, Random Forest, and XGBoost models to deliver accurate predictions, achieving a Random Forest R^2 score of 0.547. The solution aims to empower managers with data-driven insights for operational efficiency.

Objective

The primary objective is to develop a user-friendly web application that leverages machine learning to accurately predict garment worker productivity based on historical data. The app will:

- Enable managers to input features like team size, SMV, and overtime.
- Provide reliable productivity forecasts using multiple ML models.
- Enhance decision-making for production planning and resource allocation.

Scope

The project encompasses:

- Data preprocessing and feature engineering of the dataset.
- Training and evaluation of Linear Regression, Random Forest, and XGBoost models.
- Development of a basic web interface for user interaction.
- Deployment on a cloud platform (Render).

Exclusions:

- Advanced model optimization (e.g., neural networks).
- Mobile app development.
- Real-time data integration.

Problem Statement

Garment factory managers and supervisors struggle to predict worker productivity accurately due to reliance on manual estimates, which fail to account for complex factors like standard minute value (SMV), work in progress (WIP), and team dynamics. This leads to missed production targets, inefficient resource allocation, and reduced profitability, causing stress and operational challenges.

Proposed Solution

The Productivity Prediction Web App addresses these challenges through a machine learning-driven approach:

- Data Source: Utilizes the "Productivity Prediction of Garment Employees" dataset (1197 records, 14 input features: quarter, department, day, team, targeted_productivity, smv, wip, over_time, incentive, idle_time, idle_men, no_of_style_change, no_of_workers, month).

- Machine Learning Models:

- * Linear Regression: Baseline model for linear relationships ($R^2 = 0.197$, MAE = 0.107).
- * Random Forest: Ensemble model for non-linear patterns ($R^2 = 0.547$, MAE = 0.069).
- * XGBoost: Gradient boosting for complex interactions ($R^2 = 0.482$, MAE = 0.071).

- Key Features:

- * Input Interface: Users enter features via a web form (e.g., SMV, team size).
- * Model Predictions: Displays productivity forecasts from all models for comparison.
- * Preprocessing: Handles missing data (e.g., mean imputation for WIP) and encodes categorical features (e.g., quarter, department) using Scikit-learn.

- Implementation: Built with Python, Pandas, Scikit-learn, and XGBoost; web interface uses Flask; deployed on Render.

- Outcome: Provides accurate predictions to optimize production schedules and resource planning.

Resource Requirements

- Hardware:

- * Standard laptop (e.g., 8GB RAM, 2.5GHz processor) for development and model training.
- * No specialized hardware (e.g., GPU) required.

- Software:

- * Python 3.10: Core programming language.
- * Pandas, NumPy: Data preprocessing and analysis.
- * Scikit-learn: Linear Regression and Random Forest models.
- * XGBoost: Gradient boosting model.
- * Flask: Web framework for interface.
- * Gunicorn: WSGI server for deployment.
- * Visual Studio Code: Development environment.

- Personnel:

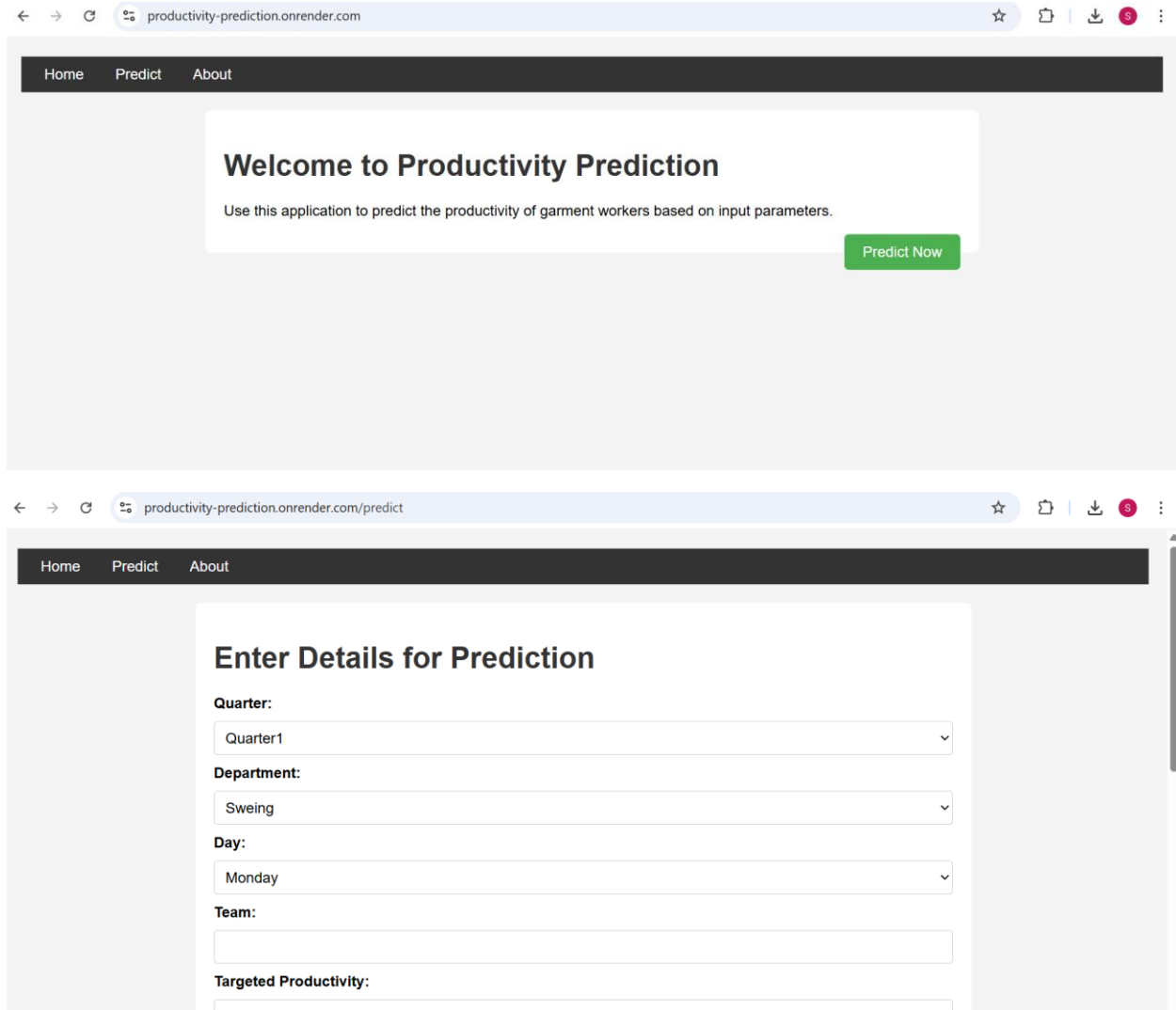
- * Solo developer with machine learning and Python expertise.
- * Estimated effort: 6 weeks, 15–20 hours/week.

- Other:

- * Dataset: Freely available from Kaggle.
- * Cloud Platform: Render (free tier) for deployment.

Images:

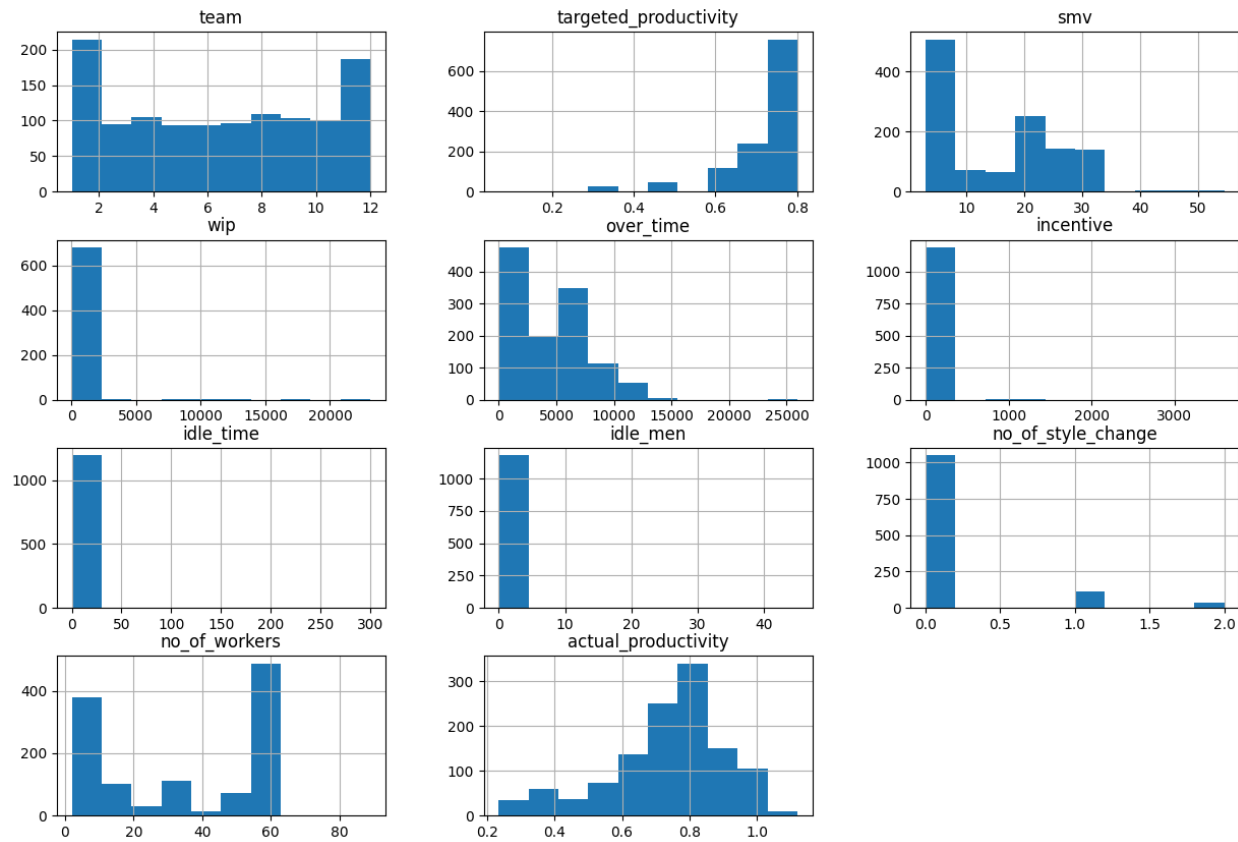
1. Figure 1: Productivity Prediction Web App Interface



- Description: Screenshot of the app's submit page, showing user inputs (e.g., SMV, team size) and predicted productivity outputs from Random Forest and XGBoost models.

- Source: Use existing file 'screenshots/submit.png' from the project directory.

2. Figure 2: Dataset Visualization



- Description: A scatter plot of actual vs. predicted productivity for the Random Forest model, highlighting model performance ($R^2 = 0.547$).

- Source: Generate using Matplotlib (script provided below) or use a stock image of a scatter plot from Unsplash (search 'data visualization').